



An Uncertainty-Guided Multi-Scale SwinUNETR Framework for 3D Larynx Segmentation in CT

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github.com/Olgameneg/larynx-swinunetr

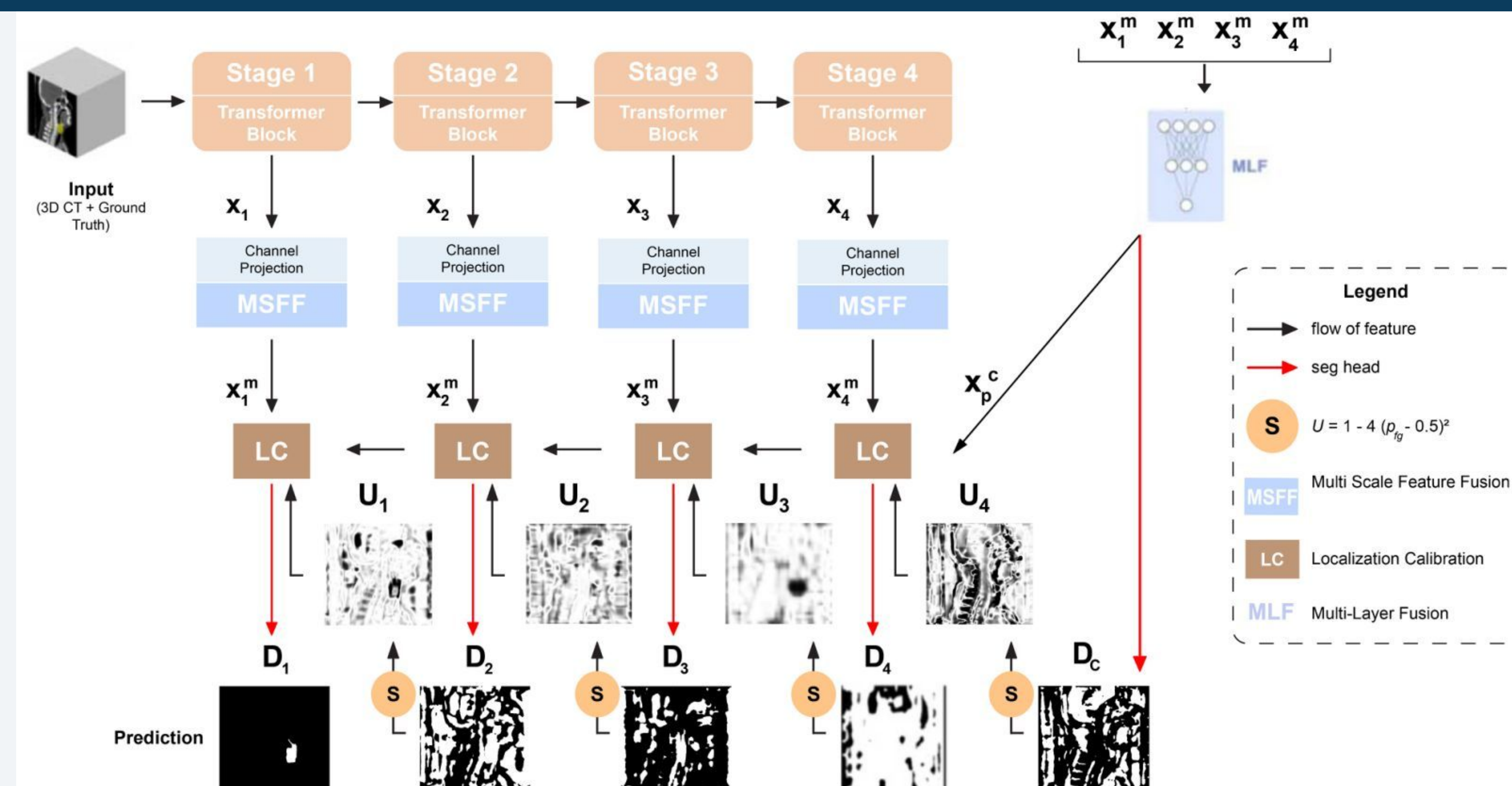
Clinical Problem

Larynx delineation on CT is mandatory for radiotherapy planning in head-and-neck cancer, and one of the hardest organs to contour:

- **Low soft-tissue contrast** blurs laryngeal boundaries.
- **Dense anatomy** in a confined region; large inter-patient variation.
- **Tumour-induced deformation** displaces healthy structures.
- **Class imbalance** — the larynx is a tiny fraction of the volume.

Most work targets healthy organs-at-risk only. **Joint segmentation of the larynx across both healthy and tumour-bearing anatomy is largely unaddressed** — precisely the setting that matters for treatment planning.

Proposed Framework



A pretrained **Swin Transformer** encoder yields features X_i ; **MSFF** refines each into X_i^m ; **MLF3D** aggregates them into the coarse prediction D_c ; a top-down cascade of **LC** modules refines it under uncertainty maps U_i . Only D_1 is supervised.

Contributions

1. Transformer framework for **whole-organ larynx delineation under oncological deformation** — healthy and cancerous anatomy in one model.
2. **Memory-efficient 3D MSFF**: three-branch ($1\times, 2\times, 0.5\times$) fusion with per-voxel softmax weights, sized for volumetric CT on a fixed GPU budget.
3. **MLF3D**: cumulative *multiplicative* fusion — a response survives only when consistent across encoder levels.
4. **Uncertainty-guided LC cascade**: voxel-wise predictive variance propagated top-down as a calibration signal — no Monte-Carlo sampling, no ensembles, fully differentiable.

MSFF3D — multi-scale fusion

$$X_i^m = X_i' \odot \bar{X}^{ms} + X_i'$$

Three branches with per-voxel softmax weights from a lightweight ConvBR fusion. Adaptive scale selection at a fraction of the memory cost.

MLF3D — cross-stage consistency

$$X_1^{mf} = X_1^m \odot X_2^m \odot X_3^m \odot X_4^m, \dots$$

Multiplicative aggregation suppresses false activations and background noise, reinforcing structures present across scales.

LC — uncertainty guidance

$$U_i = 1 - 4(p_{ig} - 0.5)^2$$

U_i modulates features via cross-attention, so **ambiguous boundary voxels receive amplified updates** while confident regions stay unchanged.

Experimental Setup

Data. *Private cohort*: 64 laryngeal-cancer + 59 healthy contrast-enhanced CT scans, manually annotated. *Public*: SegRap2023, 120 expert-annotated cases (larynx label only).

Pipeline. PyTorch + MONAI, NVIDIA A100 80 GB. Patches 128^3 ; intensities clipped to $[-200, 500]$ HU, per-volume z-score; resampled to $0.70 \times 0.56 \times 0.56$ mm.

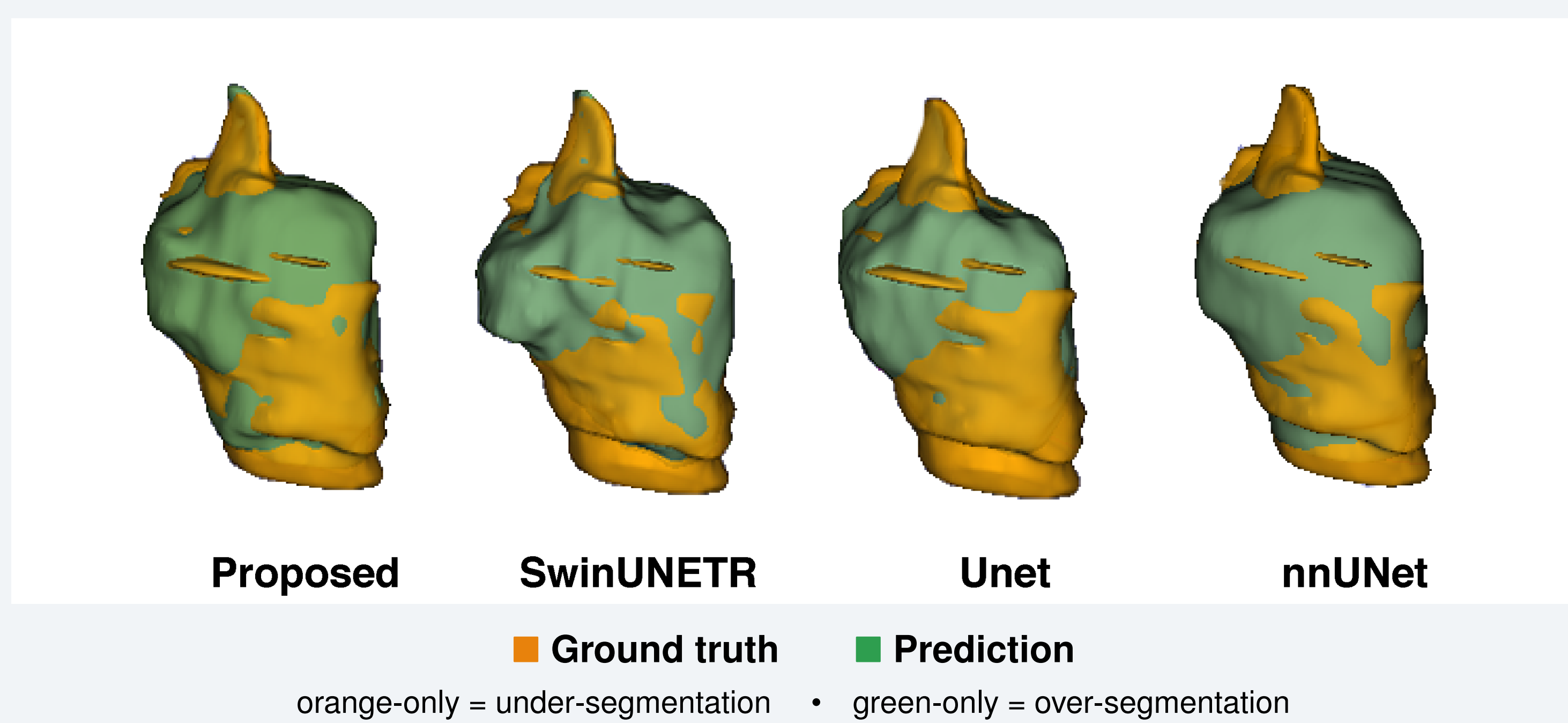
Training. AdamW, lr 10^{-4} , weight decay 10^{-5} , ≤ 1000 epochs with early stopping. SwinUNETR encoder pretrained on BTCV, feature size 48, heads (3, 6, 12, 24). $\mathcal{L} = \mathcal{L}_{Dice} + \mathcal{L}_{CE}$; sliding-window inference, **5-fold cross-validation**.

Ablation

MSFF	MLF	LC	mDice $\uparrow$	HD95 $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
—	—	—	0.8476	9.86	0.8473	0.8620
—	✓	✓	0.8193	12.15	0.8222	0.8357
✓	—	✓	0.8533	10.21	0.8753	0.8429
✓	✓	—	0.8346	11.17	0.8382	0.8443
✓	✓	✓	0.8586	9.26	0.8708	0.8494

Every module contributes. Removing **MSFF** costs the most (−3.9 Dice points), making multi-scale fusion the most critical component.

Qualitative Results — Private Cohort



Results — Private Cohort (5-fold CV)

Method	Class	mDice $\uparrow$	HD95 $\downarrow$	Prec. $\uparrow$	IoU $\uparrow$
Proposed	Cancer	0.7800	17.31	0.7780	0.6471
SwinUNETR	Cancer	0.7710	18.16	0.7573	0.6362
nnU-Net	Cancer	0.7676	17.66	0.7728	0.6358
U-Net	Cancer	0.7646	18.34	0.7707	0.6230
Proposed	Healthy	0.8851	7.22	0.8890	0.7975
SwinUNETR	Healthy	0.8804	7.59	0.8750	0.7900
nnU-Net	Healthy	0.8913	7.31	0.8960	0.8081
U-Net	Healthy	0.8821	7.70	0.8958	0.7910

Cancer: best Dice and lowest HD95 of all methods (paired t -test, $p < 0.05$ vs. every baseline) — where boundaries are hardest. **Healthy**: on par with nnU-Net (n.s.), with slightly better HD95.

Results — SegRap2023 (healthy only)

Method	mDice $\uparrow$	HD95 $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$	IoU $\uparrow$
Proposed	0.9254	2.54	0.9185	0.9346	0.8622
SwinUNETR	0.9194	2.71	0.9201	0.9207	0.8525
nnU-Net	0.9293	2.45	0.9316	0.9292	0.8696
U-Net	0.9190	2.79	0.9115	0.9300	0.8515

Competitive with nnU-Net and published SegRap2023 results (n.s.): gains on pathological anatomy cost nothing on healthy anatomy.

Conclusions

Propagating voxel-wise uncertainty as a top-down calibration signal improves larynx delineation exactly where CT is least informative — the tumour-deformed boundary — with no sampling or ensembling overhead.

- Cancer Dice **78.00%** — best of all compared methods, $p < 0.05$
- Healthy Dice **88.51%** — on par with nnU-Net, better HD95
- SegRap2023 Dice **92.54%**

Limitations & Future Work

- Pathological deformation evaluated on a **single-institution** cohort; the public dataset contains no cancerous cases.
- Transformer inference carries a **computational overhead** that may hinder clinical deployment.
- **Next**: federated multi-centre validation, topology-aware objectives, prospective dosimetric evaluation in radiotherapy workflows.